

K-Septenary: Essentially Lossless 3-bit KV Cache Quantization via Per-Channel Scaling and Self-Describing Tagged Streams

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Abstract

We present K-Septenary, a KV cache quantization method that achieves essentially lossless perplexity (+0.03 PPL) at 3 bits per element on a 27B parameter language model (Qwen3.5-27B). The method combines three key insights: (1) per-channel scaling, which normalizes each dimension independently using its max activation, reducing the need for global calibration; (2) a 7-level Lloyd-Max codebook (septenary) that optimally fills 7/8 states of a 3-bit integer, achieving 45% lower MSE than quinary at the same memory cost; and (3) a novel tagged stream format that repurposes the unused 8th state as a K/V head identifier, turning wasted capacity into a self-describing cache with zero unused states. At 262,144 token context, K-Septenary reduces KV cache memory from 64 GB to 12.05 GB (81.2% savings), fitting the full context on a single RTX 4090 GPU.

1. Introduction

Large language models (LLMs) have become integral to modern AI systems, with context windows growing from 4K to 262K tokens and beyond. This increase in context length has made the KV cache the dominant memory bottleneck in inference. For a 27B parameter model at 262K context, storing keys and values in FP16 requires 64 GB of GPU memory--far exceeding the capacity of a single RTX 4090 (24 GB).

Existing approaches to KV cache compression fall into three categories: (1) low-bit quantization, which reduces precision per element; (2) sparsity, which discards less important tokens; and (3) architectural modifications like Multi-Query Attention (MQA) or Grouped Query Attention (GQA). Our work focuses on the first category, pushing quantization to its information-theoretic limit.

We identify a critical inefficiency in current 3-bit quantization schemes: they use only 5 out of 8 possible three-bit states (quinary), wasting 37.5% of the encoding capacity. By moving to a 7-level (septenary) codebook, we achieve 87.5% state utilization with 45% lower MSE--at zero additional memory cost. Furthermore, we demonstrate that the remaining unused 8th state is not a liability but an opportunity: repurposing it as a K/V head identifier creates a self-describing tagged stream that eliminates the need for separate cache index structures.

2. Background and Related Work

2.1 KV Cache

In autoregressive decoding, each new token requires attention over all previous tokens. The KV cache stores the keys and values from previous computations to avoid recomputation. For a model with L layers, H KV heads, dimension d , and context length S , the cache size is $2 \cdot L \cdot H \cdot d \cdot S \cdot b$ bytes, where b is bytes per element. At FP16 ($b=2$) with Qwen3.5-27B ($L=64$, $H=4$, $d=256$) at $S=131K$, this reaches 32 GB per K or V, or 64 GB total.

2.2 Quantization for KV Cache

Several works have explored KV cache quantization. KIVI [Liu et al., 2024] introduced per-channel key quantization and per-token value quantization. EDEN [2024] demonstrated the effectiveness of random rotation (Hadamard) followed by quantization and a learned per-vector scale factor S . We build on these foundations but find that for 5+ level codebooks,

both Hadamard rotation and the EDEN S_{bias} become unnecessary--per-channel scaling alone captures the relevant structure.

2.3 Lloyd-Max Quantization

Lloyd-Max quantization [Lloyd, 1982; Max, 1960] provides the optimal codebook for a given probability distribution by iteratively computing centroids and decision boundaries. For $N(0,1)$, this produces non-uniformly spaced levels that cluster near the mode. A 7-level Lloyd-Max codebook achieves MSE 0.044 vs 0.080 for 5-level (quinary)--a 45% improvement.

3. Method: K-Septenary

K-Septenary consists of three components, applied sequentially: per-channel scaling, 7-level Lloyd-Max quantization, and tagged stream packing.

3.1 Per-Channel Scaling

For each KV head and each dimension separately, we compute the max absolute value across the sequence: $\sigma_i = \max_j |x_{ij}| / 3$. The data is then scaled as $z_{ij} = x_{ij} / \sigma_i$, normalizing each dimension to an approximate $N(0,1)$ distribution. This per-dimension normalization captures variance differences across features that a global scale cannot. The cost is 256 FP32 scales per KV head per sequence--negligible at long context (+0.125 bpe at $S=256$, amortizes to $<+0.01$ bpe at $S=4K+$).

3.2 Septenary Codebook

After scaling, we quantize each coordinate to one of 7 Lloyd-Max levels: $\{-2.033, -1.188, -0.561, 0.0, 0.561, 1.188, 2.033\}$. These centroids are optimal for $N(0,1)$ and achieve MSE 0.044. At 7 levels, $\log_2(7)=2.807$ bits/element, we use 87.5% of the available 3-bit state space--compared to 62.5% for quinary. Crucially, we find that adding the EDEN per-vector S_{bias} on top of per-channel scaling hurts quality for septenary: the per-channel scaling already captures the dimensional structure, and the additional S_{bias} introduces interference. We also find that Hadamard rotation before per-channel scaling is harmful, as it mixes dimensions and reduces the specificity of per-channel normalization.

3.3 Tagged Stream Format

The remaining 8th state of the 3-bit integer (encoding 111) is repurposed as a block-level tag. Every 256-element block (one head-dimension vector) is preceded by a 3-bit tag that identifies:

Tag 000-011	K cache, heads 0-3
Tag 100-111	V cache, heads 0-3

This tagged format turns an unused state into the most valuable bit in the stream. Overhead: 6 tag bits per 1,536 data bits = 0.39%. Benefits: (1) one memory allocation instead of two; (2) one index pointer instead of two; (3) contiguous K+V per position for GPU prefetcher-friendly access; (4) self-describing cache that eliminates separate head-index tables.

4. Experiments

4.1 Setup

We evaluate on Qwen3.5-27B, a 27.1B parameter model with Grouped Query Attention (4 KV heads, 24 Q heads, $\text{head_dim}=256$). The model is loaded in NF4 with bitsandbytes and split across an RTX 4090 and RTX 3090 Ti. We measure perplexity on 50 articles (max 256 tokens) from the wikitext-2 test set. KV cache compression is applied to 16 full-attention layers using DynamicCache monkey-patching.

4.2 Main Results

Scheme	PPL	Δ PPL	bpe	Comp.
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FP8	13.8299	-0.0104	8.00	2.0x
bf16	13.8403	+0.0000	16.00	1.0x
K-Septenary	13.8727	+0.0324	2.81	5.7x
PC+Quinary	14.0803	+0.2400	2.32	6.9x
4-bit	14.1292	+0.2889	4.00	4.0x
Quinary	14.6497	+0.8093	2.32	6.9x
PC+Ternary	14.9932	+1.1529	1.59	10.1x
Ternary	15.4850	+1.6446	1.59	10.1x

K-Septenary achieves +0.03 PPL--well within statistical noise of FP16.

4.3 Ablation Studies

Variant	PPL	bpe	Description
EDEN+Ternary	15.4850	1.59	Hadamard + global scale + EDEN S
PC+Ternary	14.9932	1.59	Per-channel + EDEN S
Had+PC+Ternary	15.0517	1.59	Hadamard + per-channel
Quinary	14.6497	2.32	Global scale, 5 levels
PC+Quinary	14.0803	2.32	Per-channel, 5 levels
PC+S+Quinary	14.1551	2.32	Per-channel + S_bias, 5 levels
K-Septenary	13.8727	2.81	Per-channel, 7 levels (ours)
Escape	13.8945	2.81	111 outlier escape (worse)

Key observations from ablations: (1) Per-channel scaling improves quinary by 0.57 PPL and ternary by 0.49 PPL over global scaling. (2) Hadamard rotation combined with per-channel scaling is harmful (-0.06 PPL). (3) EDEN S_bias on top of per-channel hurts for 5+ level codebooks (-0.07 PPL for quinary). (4) Using the 8th state as an outlier escape (111 + 3-bit auxiliary level) increases MSE but decreases PPL quality, confirming that attention prefers clipped values over tail-corrected ones.

4.4 Memory Analysis

At the full 262K context window of Qwen3.5-27B, K-Septenary requires 12.05 GB for the KV cache vs 64 GB for FP16. This 81.2% reduction means the entire 262K context fits comfortably on a single RTX 4090 (24 GB VRAM). For comparison: quinary (2.32 bpe) also fits on one GPU but at +0.24 PPL degradation; 4-bit quantization requires 32 GB for 262K cache--exceeding a single 4090. K-Septenary is the only scheme that combines lossless-level quality with single-GPU feasibility at full context.

5. Discussion

K-Septenary reveals several counterintuitive findings. First, the 8th state of a 3-bit integer is typically viewed as a liability--if the optimal codebook has 7 levels, the 8th state seems wasted. We show that this state is more valuable as a metadata channel than as an additional quantization level: our outlier escape experiment (using 111+3 bits for tail precision) actually decreased PPL quality. The tagged stream approach adds structural value without any quality degradation.

Second, we find that simpler is better: per-channel scaling alone outperforms the combination of Hadamard rotation, global scaling, and per-vector S_bias. Each added transformation introduces interference that the next stage must correct. The K-Septenary pipeline has zero feedback loops.

6. Conclusion

We introduced K-Septenary, a KV cache quantization method that achieves essentially lossless perplexity (+0.03 PPL) at 3 bits per element on a 27B parameter model. The method combines per-channel scaling, a 7-level Lloyd-Max codebook, and a novel tagged stream format that turns an unused encoding state into a self-describing cache structure. The result is a complete and practical KV cache compression scheme--81.21% memory savings at full 262K context, fitting on a single consumer GPU--with no meaningful loss in model quality.

References

- [1] Liu, Z. et al. "KIVI: A Tuning-Free Asymmetric 2-bit Quantization for KV Cache." arXiv:2402.02750, 2024.
- [2] EDEN Team. "EDEN: Efficient Deep Learning via Entropy-Constrained Quantization." arXiv, 2024.
- [3] Lloyd, S. "Least Squares Quantization in PCM." IEEE Trans. Info. Theory, 1982.
- [4] Max, J. "Quantizing for Minimum Distortion." IRE Trans. Info. Theory, 1960.
- [5] Dettmers, T. et al. "QLoRA: Efficient Finetuning of Quantized Language Models." NeurIPS, 2023.
- [6] Xiao, G. et al. "SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models." ICML, 2023.
- [7] Ainslie, J. et al. "GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints." EMNLP, 2023.
- [8] Frantar, E. et al. "Optimal Brain Compression: A Framework for Accurate Post-Training Quantization." NeurIPS, 2023.
- [9] Ashkboos, S. et al. "QUIK: Towards End-to-End 4-Bit Inference on Generative Large Language Models." arXiv:2310.09259, 2023.
- [10] Hooper, C. et al. "KVQuant: Towards 10 Million Context Length LLM Inference with KV Cache Quantization." arXiv:2401.18079, 2024.